

MAJOR INCIDENT REPORT — INC0001234

Priority: P1 | Category: Network Infrastructure | Status: Resolved
Opened: 2025-01-14 03:47 UTC | Resolved: 2025-01-14 06:22 UTC | Duration: 2h 35min
Incident Commander: James Wu (CTO) | SLA Outcome: BREACHED by 35 minutes

1. Executive Summary

A BGP routing outage severed APAC connectivity for 847 users across Singapore, Sydney, and Tokyo on 14 January 2025 at 03:47 UTC. Ansible Tower job AT-4421 erroneously applied a disaster-recovery BGP community filter to three production core routers, withdrawing all peer advertisements to upstream ISPs. A logic error in the CI/CD pipeline approval gate allowed the automation job to bypass the required Network CAB review. Manual intervention on router APAC-CORE-R3 delayed full recovery. Total business impact: USD 142,000. P1 SLA breached by 35 minutes.

2. Incident Timeline

Time (UTC)	Event	Actor	Tool / System	Outcome
03:41	AT-4421 automation job executes	Ansible Tower pipeline	Ansible Tower	BGP DR policy applied to prod routers APAC-CORE-R1, R2, R3
03:47	BGP peer-loss alert fires	AutoMonitor	Grafana + PagerDuty	3 peers DOWN; P1 INC0001234 auto-created; on-call paged
03:51	On-call engineer acknowledges	Sarah Chen	PagerDuty	Bridge #inc-1234 opened; runbook RB-NET-014 retrieved
03:58	BGP routing table inspected	Sarah Chen	SSH to APAC-CORE-R1	All 3 routers show peers in Idle state; no hardware faults
04:02	Change AT-4421 correlated	Sarah Chen + NOC	ServiceNow Change Calendar	Job ran 6 min before alert; flagged as root cause
04:09	ECAB convened; rollback approved	James Wu (CTO)	Slack bridge + phone	ECAB ref EC-2025-003 issued; Mike Torres leads rollback
04:15	Rollback on R1 and R2 succeeds	Mike Torres	Ansible Tower	BGP peers on R1/R2 re-established within 90 seconds
04:31	R3 unresponsive to automation	Mike Torres	SSH direct to APAC-CORE-R3	Timeout on automation; manual BGP config revert applied
04:45	72 % users restored	NOC	Grafana	APAC-E2 edge still showing 4 % packet loss; investigation continues
05:10	Residual static route found on APAC-E2	Sarah Chen	SSH to APAC-E2	Static route remnant from AT-4421 blocking /24 prefix; removed
05:22	Full BGP convergence achieved	Network Team	Grafana BGP dashboard	All 6 peers Established; route table verified vs baseline
06:22	P1 incident closed	Incident Manager	ServiceNow	All services confirmed normal; post-mortem booked 2025-01-21

3. Business Impact

Impact Area	Detail	Est. Cost
SAP ERP transactions	3,412 EDI transactions queued; 47 failed and required manual replay after restoration	USD 94,000

Salesforce CRM	847 APAC users locked out 2h 35min; 47 active deals in Singapore and Sydney affected	USD 28,000
Staff productivity	847 users × 2.5 hr average × \$28/hr blended cost	USD 48,000
Customer SLA credits	Acme Corp, TechNova, BlueWave raised SLA claims — credits estimated	USD 12,000
Incident response overtime	6 engineers × 4 hr × \$85/hr	USD 2,040
TOTAL		USD 142,000

4. Root Cause Analysis

4.1 Immediate cause

Ansible Tower template *network-bgp-policy-sync* resolved its variable file path using a misconfigured inventory group variable. Git commit **a3f9c12** by engineer Ravi Patel on 2025-01-13 overrode the production variable *bgp_policy_vars_file* with *vars/bgp_dr_policy.yml* instead of *vars/bgp_prod_policy.yml*. The DR policy applies community 65100:999 — a BGP blackhole tag that instructs all three upstream ISPs to withdraw the routes.

4.2 Contributing cause — CI/CD approval gate defect

Pipeline version 2.14 contained a conditional logic error: it evaluated *change_state != 'Cancelled'* rather than *change_state == 'Approved'*. Change CHG0009823 was in *Scheduled* state — not CAB-approved — yet the job proceeded. This defect was present for 6 weeks and affected all network automation runs since version 2.14 was deployed.

5. Corrective Actions

Ref	Action	Owner	Due	Status
CA-01	Fix CI/CD gate: evaluate state == Approved (PRB0003312)	Platform Eng.	2025-01-21	In Progress
CA-02	Add pre-flight diff check: abort if var file env tag mismatches inventory	Automation Team	2025-01-28	In Progress
CA-03	Mandatory second-engineer sign-off for all prod BGP automation jobs	Network Ops Lead	2025-01-18	Complete
CA-04	BGP peer-count synthetic alert: P1 if fewer than 6 peers for 90 seconds	NOC	2025-01-20	Complete
CA-05	Update runbook RB-NET-014 with APAC-E2 static route cleanup steps	Sarah Chen	2025-01-24	Pending
CA-06	Audit all 47 existing Ansible Tower network templates for similar variable override risks	Automation Team	2025-02-07	Pending
CA-07	Post-mortem with APAC leads and affected customer account managers	Incident Manager	2025-01-21	Scheduled